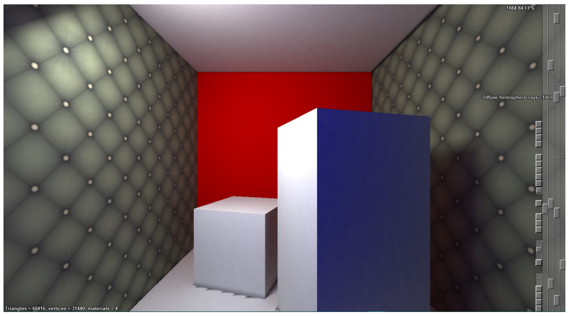
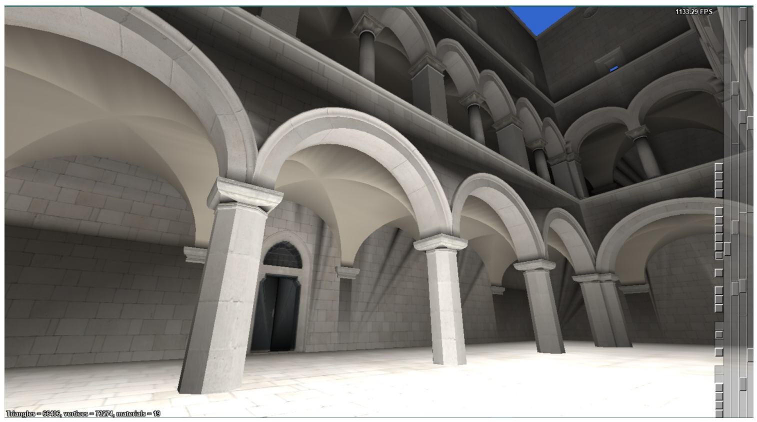
Syed Ashraf

Portfolio 2025

github.com/Cy5er/Portfolio

Advanced Computer Graphics Assignments





- This project focused on extending an accelerated ray tracing framework by incorporating diffuse global illumination using a simplified radiosity technique. The implementation involved calculating direct lighting from area light sources, simulating indirect light bounces iteratively, and rendering the final result using vertex-based radiosity to achieve effects like color bleeding.
- Key techniques used included a bounding volume hierarchy with surface area heuristic for efficient ray intersection, quasi-Monte Carlo sampling of area lights for direct irradiance, and an iterative radiosity solver to approximate indirect lighting.
- A major challenge was achieving visually accurate global illumination while maintaining good performance. Balancing computational cost with visual quality was critical for producing convincing results.
- Detailed Overview

Evaluating Language Models for Generating and Judging Programming Feedback

You are a CS professor teaching introductory programming using Python.

Below are a problem description, test cases, and an incorrect program written by a student (i.e., it does not pass all test cases). You are also provided with the ground truth description of a bug in that program and the required fixes.

<description>, <test cases>, <code>, <bug description>, <bug fixes>

Below is a list of bugs and their fixes written by a teaching assistant.

<LLM feedback generation>

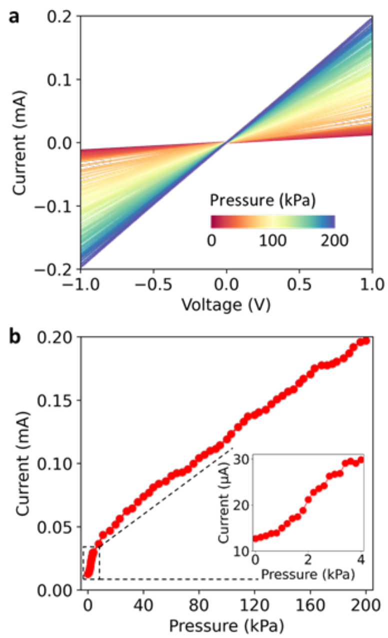
Your task is to evaluate the quality of the TA's feedback according to the grading criteria outlined below, using the provided ground truth bug description and fixes as reference.

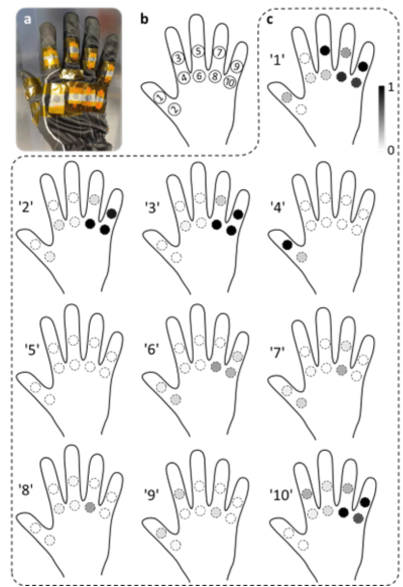
...

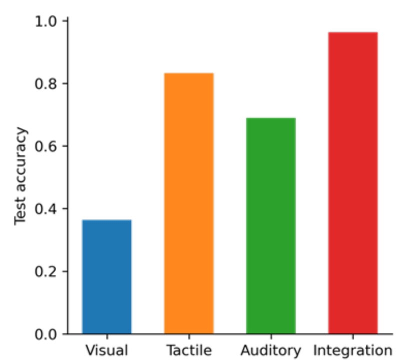
model	EA	ES	EC	FA	FS	FC	E_{all}	F_{all}	ALL
Gemma-2b	0.44	0.02	0.65	0.42	0.02	0.65	0.00	0.02	0.00
Phi-3-mini	0.72	0.19	0.93	0.74	0.21	0.89	0.18	0.19	0.18
Mistral-7b	0.70	0.23	**0.96**	0.70	0.25	0.93	0.21	0.21	0.19
Llama3.1-8b	0.70	0.04	0.74	0.68	0.05	0.72	0.04	0.04	0.04
Llama3.1-70b	0.89	0.28	0.89	0.88	0.40	0.89	0.26	0.37	0.26
GPT-3.5-turbo	0.84	0.35	0.93	0.81	0.37	0.84	0.32	0.28	0.28
GPT-4o-mini	0.96	0.35	0.93	0.93	0.37	0.96	0.30	0.35	0.30
GPT-4o	**0.98**	**0.44**	0.93	**0.93**	**0.42**	**0.98**	**0.39**	**0.42**	**0.39**

- This paper evaluates how large language models, both proprietary and open source perform, in generating feedback for programming assignments and in judging the quality of such feedback. The paper uses a benchmark dataset of student Python programs and evaluates LLMs across two core tasks: generating explanations and fixes for bugs, and evaluating feedback quality.
- The main challenge was assessing how reliably LLMs can generate helpful feedback on top of serving as evaluators of others' feedback. The work addressed concerns over the trustworthiness and consistency of model judgments, particularly regarding identifying only real issues without hallucinating bugs.
- I contributed as a co-author, participating in experimentation and evaluation of language model performance, including grading consistency and model comparisons.
- arXiv.

Neuromorphic Multisensory Numerosity Perception Enhanced by a Tactile Glove

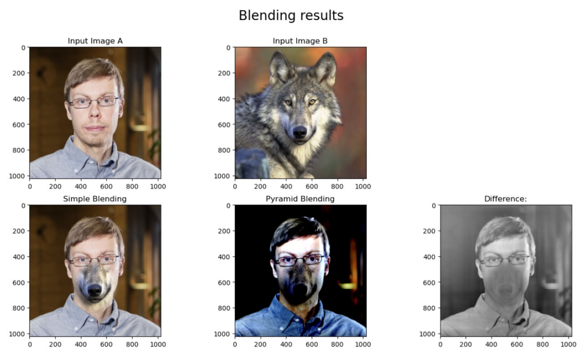


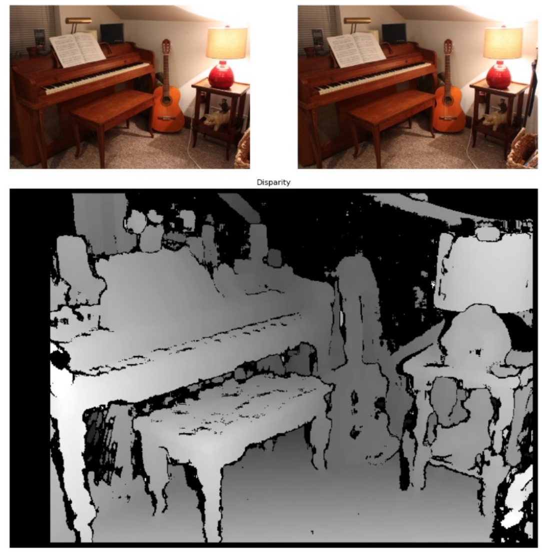




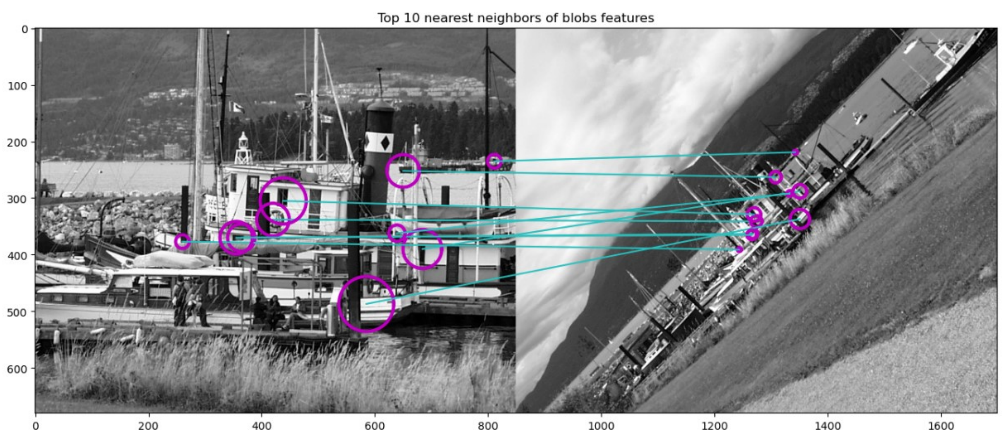
- This paper presents a neuromorphic tactile glove enhanced with MXene-based pressure sensors, enabling gesture recognition for improved numerosity perception. The glove detects finger joint movement to recognize hand gestures representing numbers, which are then integrated with vision and auditory signals to emulate multisensory processing similar to the human brain. The approach aims to enhance cognitive perception tasks such as quantity estimation in robotics and human-machine interaction.
- The main technical contributions include the design of high-sensitivity pressure sensors, the use of very simple neural networks for classifying American Sign Language numbers, and to show how the fusion of tactile, visual, and auditory inputs to improve recognition accuracy despite really simple models.
- I contributed as a co-author, developing the dataset, in building and evaluating the multimodal models, and in creating and evaluating the performance of the glove.
- Paper Abstract

Computer Vision Assignments



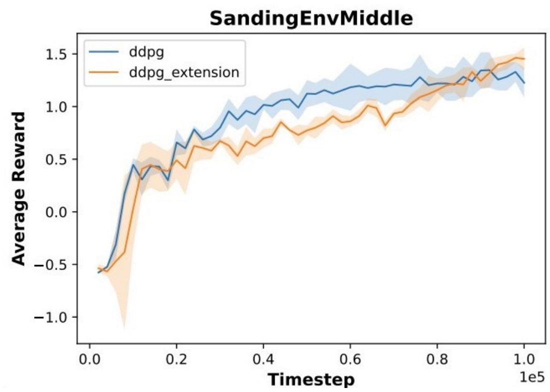


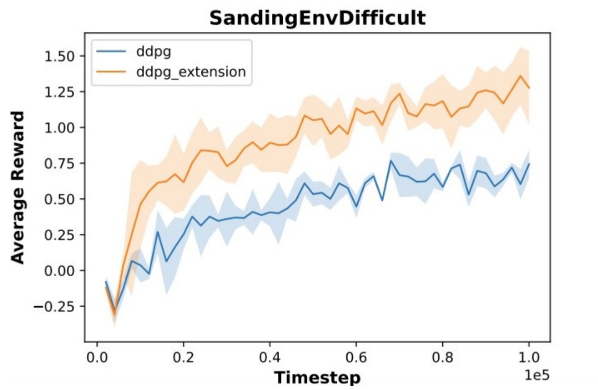
Blob detection and matching



- These assignments focus on the building blocks of computer vision such as image processing, feature extraction, object recognition, and the integration of deep learning methods. From low-level operations to high-level inference tasks, providing a well-rounded understanding of computer vision tasks.
- A major challenge involved bridging traditional computer vision techniques with deep learning models. In particular, integrating feature descriptors with learned representations for more robust detection and recognition. Tuning models and understanding when to use each technique effectively.
- Detailed Overview

Reinforcement Learning Sanding Project

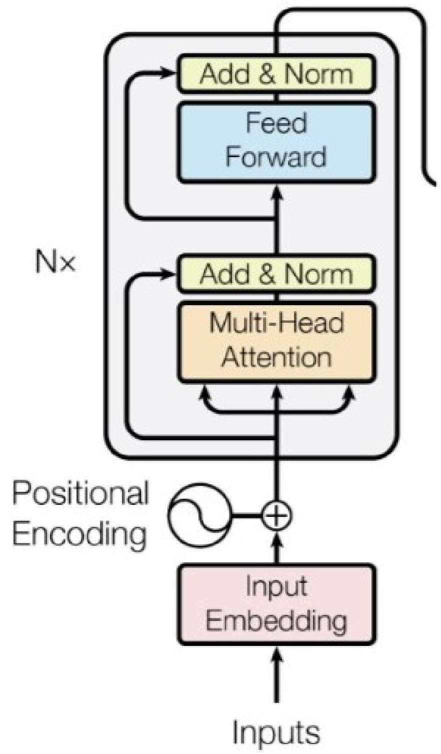




- This project focuses on training a reinforcement learning agent to control a sanding robot in a simulated 2D environment. The robot learns to maximize reward by sanding green zones and avoiding red painted areas. We implemented and evaluated both a standard DDPG algorithm and an extended version incorporating N-step bootstrapping (LNSS).
- The full pipeline includes custom environment construction, visualization, actor-critic network implementation, experience replay, and agent evaluation across multiple difficulty levels. Videos of the performance of the model can be found via the link at the top of the page.
- We implemented separate base DDPG algorithms it was a part of a previous assignment. We wrote the extended DDPG algorithm collaboratively and the visualization tools were provided by the course staff.
- Detailed Overview

SNLP Toxicity Project

Toxic text output
<|startoftext|><|1|> they should all be held to account......
<|startoftext|><|1|> what a stupid comment. if i had that much my children would kill me. a couple bucks that would have to pay for something they didn't need.
<|startoftext|><|1|> it is not just the press and the liberal media who hate trump. it is the political class of America that hates the media and this is the key to destroying the country
Nontoxic text output
<|startoftext|><|0|> they should all be held to account......
<|startoftext|><|0|> maybe he should be more than just an admitted racist.
<|startoftext|><|0|> it is not just the government that does not have the resources to manage. it is the community that has to step up and take responsibility for managing the community. so much so that the residents are forced to have their elected representatives look after their citizens and do what is right for them. there is an ongoing effort to reduce the burden of the people and make people feel like something exists no matter what. it s not about them not them. it s about them.



Example	Test data input	Tokenized
1	I couldn't disagree more with this column; Canadians have moved on and don't care...	[CLS, disagree, more, with, this, have, moved, on, and, care,...]
2	I get the odd feeling Klastri the head of the ACLU of...	[CLS, get, the, odd, feeling, ", the, head, of, the, of, ",...]
3	"Had"... and is now outed as a hypocrite.	[CLS, ..., "had", ..., and, ..., is, now, outed, as, a, EOS]

Model	Changes	Performance
Baseline	None	Acc: 64%, F1: 0.00
LR Variation	lr = 0.01	Acc: 64%, F1: 0.00
LR Variation	lr = 0.001	Acc: 92%, F1: 0.89
LR Variation	lr = 0.0001	Acc: 84%, F1: 0.80
LR & MLP Variation	lr = 0.001, 8 * num_features	Acc: 88%, F1: 0.84

- This project focused on developing a toxicity classification system as part of the Statistical Natural Language Processing course. The aim was to distinguish toxic from non-toxic comments and explore text detoxification through language generation.
- We implemented a transformer-based encoder combined with an MLP classifier, and explored vocabulary-level tokenization, embedding layers, and masked training.
- My contributions included general code debugging, getting the code to work and deployed on the Aalto Triton cluster with GPUs. I was responsible for running all the models along with hyperparameter optimization, running model predictions and checking model performance.
- Project Report